

**Beyond the Family: How Neighborhood Poverty Academic Outcomes Across
U.S. School Districts**

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Introduction

Many people assume that a child's academic success is mostly dependent on their family's economic status. This is because they believe that higher income families have better resources, more stable housing and more access to quality enrichment opportunities that facilitate academic success. This is true for the most part, but this study hopes to offer a different lens to view academic success through, by considering the broader neighborhood context where children grow up.

This report is prepared for consideration by state education commissioners who are faced with the difficult task of allocating resources to the thousands of school districts with different socioeconomic profiles. The common policy is to give the most resources based on family poverty level, but this strategy may be more effective if it was amended to take into account the poverty level and socio economic conditions of the surrounding neighborhood. A child from a low income family that lives in a stable, well resourced neighborhood may outperform a child from a middle income family in a high-poverty community with little to no resources. In other words, community conditions can both increase and decrease the impact of individual family circumstances on academic outcomes.

For the following analysis, data from the Standard Education Archive Version 6.0 (SEDA 6.0) was used. This is a large and informative dataset that covers 450 million standardized test scores over 12,000 U.S. public school districts from 2008 - 2019. It is used to examine the relationship between neighborhood level poverty, family income, district urbanity and average academic achievement at the district level. Both frequentist and Bayesian statistical methods are used to assess the strength, direction and uncertainty of these relationships.

Descriptive Overview

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Variables and Summary Statistics

This analysis focuses on four variables from the Stanford Education Data Archive Version 6.0 (SEDA 6.0), and covers 12,335 U.S. public school districts across 2008-2009 through 2018-2019 school years. The dataset has comprehensive coverage of the national population of public school districts which allows for population-based analysis and inference.

Summary Statistics for All Analytical Variables (n = 12,335 districts)

Variable	Min	Mean	Median	Max	Distribution
Test Score (Cohort Scale)	-2.438	0.008	0.007	1.331	Approx. Normal
Family Income (Log)	9.838	10.859	10.819	12.334	Approx. Normal
Neighborhood Poverty Rate	0.007	0.130	0.122	0.472	Right Skewed
Urbanicity — City	n = 687 (5.6%)				
Urbanicity — Suburb	n = 2,809 (22.8%)				
Urbanicity — Town	n = 2,298 (18.6%)				
Urbanicity — Rural	n = 6,541 (53.0%)				

Note. Test scores are on the SEDA cohort scale where 0 = national average and 1 unit = 1 standard deviation. Family income is log-transformed. Neighborhood poverty rate is the proportion of children below the federal poverty line. Urbanicity is a categorical variable; City is the reference category in all regression models.

To explore whether neighborhood poverty behaves differently across income groups, the family income variable was divided into three categories. High Income (log income above 11.03), Middle Income. And Low Income, (log income below 10.65). This variable is used for interaction analysis, not as a substitute for family income.

Outcome Variable

Test Score

This averages district test scores on the cohort scale. It is centered near zero (mean = 0.08, median = 0.007) and it is almost normally distributed. This can be confirmed by the histogram in Figure 1. On the cohort scale, scores are calculated so that zero represents the national average and one unit equals one standard deviation. There is a slight skew to the left in the histogram, which tells us that there is a minority of districts that score substantially below the national average.

The fact that there is a range of almost 4 standard deviations in test scores across school districts provide evidence that there is an inequality of academic outcomes across the United States. The best performing district scored more than 1.3 above the mean while the worst performing district scored 2.5 below the national standard. This spread shows an inequality this analysis hopes to explain.

Predictor Variables

Family Income

Median household income is represented in log form and was provided by the dataset. The log transformation was applied due to the fact that income distributions are naturally right-skewed. In log form, we get a normally distributed variable as can be seen in Figure 2. The range of values spans from 9.838 and 12.334 in log amounts and \$18,732 to \$227,294 in dollar amounts, which captures the economic conditions in districts across the United States.

Outliers are not an issue for this variable because extreme variables on both high and low ends represent wealthy and low-income rural and urban areas that are important for analysis.

Neighborhood Poverty Rate

The neighborhood poverty rate represents the proportion of children in the district living below the federal poverty line. There is a right skewed distribution shown in Figure 3. Most districts

cluster between poverty rates of 0.07 and 0.20 but the tail extends to a maximum of 0.472, telling us that almost half of all children in some districts live in poverty.

There is a distinction to be made between this variable and family income. While the two variables are correlated ($r=-0.68$), they are not the same thing. A district can have a median income but a large concentration of children in poverty, if there is unequal income distribution in a neighborhood. The scatterplot of family income vs. neighborhood poverty (Figure 4) shows this as the points around the regression line shows that the two variables capture distinct dimensions of economic disadvantage, even though there is a clear negative relationship. This supports the decision to keep both neighborhood poverty rate and family income in the regression model.

Pearson Correlation Results — All Students (n = 12,335)

Predictor vs. Test Score	r	t	df	p-value	95% CI
Family Income	0.677	102.02	12,333	< 2.2e-16	[0.667, 0.686]
Neighborhood Poverty	-0.684	-104.09	12,333	< 2.2e-16	[-0.693, -0.674]

Note. Both correlations are statistically significant at $p < 0.001$. Neighborhood poverty shows a slightly stronger correlation magnitude than family income, providing initial evidence that neighborhood-level conditions are at least as predictive of academic outcomes as family-level income.

Urbanicity

The urbanicity variable classifies each district into 4 geographic categories. They are City, Suburb, Town, and Rural, as shown in Figure 5. There is a large representation of rural districts in the dataset, but this reflects the population as there are more rural districts than urban, even though urban areas have more population density, City is used as the reference category in the regression model.

The test scores by urbanicity boxplot (Figure 6) reveals that suburban districts have the highest median scores and the tightest range of values, while city districts have the largest spread. The

neighborhood poverty by urbanicity boxplot (Figure 7) shows that city districts have higher median poverty rates than suburban districts, but rural districts have the lowest poverty rates overall. Some rural districts are very wealthy while others are substantially impoverished as can be seen from the outliers. This would be important for the commissioner because this heterogeneity means that rural districts are not meant to be treated as a homogenous group.

Outliers and Data Quality

Outliers are present at the extremes of all the continuous variables in the dataset. Since this dataset represents the near-complete population of U.S. public school districts rather than a sample, these outliers were not removed.

One data quality limitation is that 378 districts (approximately 3% of the original 12,713) were dropped due to missing values in our variables. These represented very small districts that SEDA had difficulties getting reliable information from.

Inferential Analysis

Predictor and Response Variables

The response variable in all models is the Test Score, representing the mean academic achievement across grades 3-8 in math and English/Language Arts, averaged across the 2009 to 2019 period. The three predictor variables included in the model are family income, neighborhood poverty rate, district urbanicity with City as the reference level.

The final model was selected after evaluating multicollinearity using Variance Inflation Factors (VIF). The initial model included the SES index which was calculated using the neighborhood poverty rate, resulting in a large VIF value. Removing the SES index variable reduced our VIF values to more acceptable levels as seen in the table below. Urbanicity was retained because it provides a categorical predictor that adds geographic context without contributing to multicollinearity,

Variance Inflation Factor (VIF) Diagnostics — Base Model

Predictor	GVIF	Df	GVIF ^{1/(2*Df)}
Family Income	4.145	1	2.036
Neighborhood Poverty	3.749	1	1.936
Urbanicity	1.442	3	1.063

Note. GVIF = Generalized Variance Inflation Factor. Values below 5 indicate acceptable multicollinearity. An initial model including the SES composite index (sesavgall) produced VIF values of 9.69 and 7.05 for ses_index and nbhd_poverty respectively, indicating severe overlap. The SES index was removed and replaced with the categorical urbanicity variable, bringing all VIFs within acceptable range.

Research Question 1: Does Neighborhood Poverty Predict Academic Outcomes

Null Hypothesis (H_0): After controlling for family income and district urbanicity, neighborhood poverty rate has no significant independent effect on the average test scores in districts.

Alternate Hypothesis (H_1): After controlling for family income and district urbanicity, neighborhood poverty rate has a significant and negative effect on district average test scores.

The frequentist analysis was the first analysis. It determined that family income correlates positively with test scores ($r = 0.677$, 95%CI[0.667, 0.686], $t=102.02$, $p < 2.2e-16$), confirming something already known and neighborhood poverty correlated negatively with test scores. The magnitude of the neighborhood poverty correlation is slightly larger than that of family income, hinting at the fact that neighborhood level conditions are as important as family-level income when predicting academic success.

The full multiple linear regression model explains 50.6% of the variance in district test cores ($R^2=0.506$, $F(5, 12329) = 2527$, $p < 2.2e-16$). Both key predictors are significant as shown in Table 2.

OLS Regression Results — Base Model

Outcome: District Average Test Score | $R^2 = 0.506$ | $F(5, 12329) = 2527$ | $p < 2.2e-16$

Predictor	Estimate	Std. Error	t-value	p-value
(Intercept)	-3.649	0.157	-23.243	< 2e-16 ***
Family Income (log)	0.362	0.014	26.604	< 2e-16 ***
Neighborhood Poverty	-2.137	0.065	-32.863	< 2e-16 ***
Urbanicity: Suburb	0.005	0.011	0.498	0.618
Urbanicity: Town	0.031	0.011	2.884	0.004 **
Urbanicity: Rural	-0.006	0.010	-0.554	0.580

Note. Reference category for urbanicity is City. Residual standard error = 0.2391. *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$.

The neighborhood poverty coefficient tells us that a 10% point increase in neighborhood child poverty is associated with a 0.214 standard deviation decrease in test scores after controlling for family income and urbanicity. We can reject the null hypothesis here as neighborhood poverty has a strong and statistically significant negative effect on academic outcomes that is independent of family income.

The Bayesian analysis reinforces these results. The posterior distribution for the neighborhood poverty coefficient (Figure 8) is concentrated below zero. The posterior mean is -2.137 and the 95% credible interval is [-2.261, -2.010]. This allows for the statement that there is a 95% probability that the true effect of neighborhood poverty on test scores falls between -2.261 and -2.010 standard deviations per unit.

Bayesian Posterior Estimates and 95% Credible Intervals Base Model

MCMC Sampling: 10,000 iterations | Convergence confirmed via trace plot inspection

Parameter	Post. Mean	Post. SD	2.5%	97.5%
Family Income	0.362	0.014	0.335	0.389
Neighborhood Poverty	-2.137	0.065	-2.265	-2.010
Urbanicity: City	-0.007	0.007	-0.021	0.007
Urbanicity: Suburb	-0.002	0.005	-0.012	0.007
Urbanicity: Town	0.023	0.005	0.014	0.032
Urbanicity: Rural	-0.013	0.004	-0.021	-0.006

Note. Posterior mean and standard deviation derived from 10,000 MCMC samples. 2.5% and 97.5% columns represent the lower and upper bounds of the 95% credible interval. A credible interval entirely below or above zero indicates a parameter whose effect can be stated with 95% probability to be negative or positive respectively. The neighborhood poverty credible interval [-2.265, -2.010] is entirely negative and exceeds 1 in absolute magnitude, indicating a large and certain negative effect on academic outcomes.]

The family income posterior (mean = 0.362, 95% CI[0.335, 0.389]) is also well-defined and positive. The posterior distributions plot (Figure 9) shows the effects side by side. The MCMC trace plots for both parameters show the fuzzy caterpillar pattern that indicates convergence, further attesting to the reliability of the posterior estimates.

Research Question 2: Does Urbanicity Moderate the Neighborhood Poverty Effect?

The urbanicity coefficients in the regression reveal something interesting. Using City districts as the reference category, Suburb districts score 0.005 standard deviations higher(not significant, $p = 0.0618$), Town districts score 0.031 standard deviations higher (significant, $p = 0.004$), and

Rural districts score 0.006 standard deviations lower (not significant, $p = 0.580$). The small magnitude and presence of insignificance of the urbanicity coefficients is surprising considering the visual difference in test scores.

Once neighborhood poverty and family income are controlled for, the residual difference in test scores that can be attributed to test scores is small. This means that the disadvantage associated with urban and rural districts compared to suburban ones is not due to geographic factors but is mostly explained by the higher poverty rates and lower incomes within these districts. Figure 10 shows this as within each urbanicity category, the negative relationship between poverty and test scores has a similar slope. A high poverty rural district will score comparably to a high poverty city district.

The significant urbanicity effect of Town districts compared to City districts could be due to a structural advantage in education systems in small towns or their culture when it comes to community living.

The Overall Model

Our full model (Test Score $\sim$ Family Income + Neighborhood Poverty Rate + Urbanicity) explains 50.6% of the variance in the district test scores ($R^2 = 0.506$). The most powerful predictors are Family Income and Neighborhood Poverty Rate, while Urbanicity adds a small explanatory power.

Supplementary Analysis: Do Interaction Effects Exist?

Income Group Interaction

OLS Regression Results — Income Group Interaction Model

Outcome: District Average Test Score | $R^2 = 0.532$ | $F(6, 12328) = 2332$ | $p < 2.2e-16$

Predictor	Estimate	Std. Error	t-value	p-value
(Intercept)	-3.547	0.198	-17.930	< 2e-16 ***
Family Income (log)	0.370	0.017	21.687	< 2e-16 ***
Neighborhood Poverty (High Income ref.)	-4.854	0.151	-32.104	< 2e-16 ***
Income Group: Middle Income	-0.083	0.021	-3.986	< 0.001 ***
Income Group: Low Income	-0.214	0.020	-10.551	< 2e-16 ***
Poverty × Middle Income	1.724	0.185	9.335	< 2e-16 ***
Poverty × Low Income	3.050	0.151	20.172	< 2e-16 ***

Note. Reference category is High Income (top third of family income distribution). Poverty × income group interaction terms represent the degree to which the poverty slope is moderated by income group relative to High Income districts. Residual standard error = 0.2329. *** $p < 0.001$.

The first interaction model ($R^2 = 0.532$ which is up from 0.506 in the base model) presents an interesting finding. The base neighborhood coefficient for High Income districts is -4.854, which is large and negative. This is twice the coefficient of that in our base model. This provides evidence that poverty in high income districts is associated with a steep decline in academic performance. However, the interaction terms of neighborhood poverty and middle income families (coefficient = 1.72373, p-value = < 2e-16 ***) and neighborhood poverty and low income (coefficient = -3.04993, p-value = < 2e-16 ***) tells us that each additional unit of poverty does marginally less damage as district income decreases. Communities with conditions of lower income have learned to function in poverty conditions. The Bayesian intervals confirm this pattern. We are 95% confident the High Income interaction falls between [-1.788, -1.381] and the Low Income interaction falls between [1.310, 1.600]. Both intervals do not include zero which is proof of their credibility.

Bayesian Posterior Estimates — Income Group Interaction Model

Parameter	Post. Mean	Post. SD	2.5%	97.5%
Family Income	0.369	0.017	0.336	0.403
Neighborhood Poverty	-3.258	0.076	-3.409	-3.110
Income Group: Middle Income	0.033	0.004	0.025	0.041
Income Group: Low Income	0.074	0.005	0.063	0.085
Poverty × High Income	-1.583	0.104	-1.788	-1.381
Poverty × Middle Income	0.127	0.096	-0.064	0.314
Poverty × Low Income	1.456	0.074	1.310	1.600

Note. We are 95% confident the High Income interaction effect falls between -1.788 and -1.381, and 95% confident the Low Income interaction effect falls between 1.310 and 1.600 — both intervals sitting entirely away from zero. The Middle Income interaction credible interval [-0.064, 0.314] crosses zero, indicating uncertainty about the direction of this effect.

Urbanicity Interaction

OLS Regression Results — Urbanicity Interaction Model

Outcome: District Average Test Score | $R^2 = 0.518$ | $F(8, 12326) = 1657$ | $p < 2.2e-16$

Predictor	Estimate	Std. Error	t-value	p-value
(Intercept)	-3.209	0.159	-20.187	< 2e-16 ***
Family Income (log)	0.313	0.014	22.794	< 2e-16 ***
Neighborhood Poverty (City ref.)	-1.646	0.127	-12.917	< 2e-16 ***
Urbanicity: Suburb	0.225	0.024	9.458	< 2e-16 ***
Urbanicity: Town	0.050	0.026	1.942	<u>0.052</u>
Urbanicity: Rural	0.085	0.024	3.591	< 0.001 ***
Poverty × Suburb	-1.754	0.139	-12.622	< 2e-16 ***
Poverty × Town	-0.100	0.140	-0.715	0.475
Poverty × Rural	-0.538	0.129	-4.180	< 0.001 ***

*Note. Reference category for urbanicity is City. Poverty × urbanicity interaction terms represent how much the poverty slope changes relative to City districts. Residual standard error = 0.2363. *** $p < 0.001$, $p < 0.10$.*

The urbanicity interaction tells the same story from a different angle. This model has a fit of 51.8%, which is also an improvement on our base model. Suburban districts, the most affluent

and resource rich district type, have the steepest drop in academic performance when neighborhood poverty is present. Rural districts show a smaller but significant penalty of -0.538, while town districts show no meaningful difference from city districts ($p = 0.475$). The Bayesian credible interval for the suburban interaction is $[-1.285, -1.021]$, which does not include zero, confirming this is a credible effect. Looking at them together, both interactions point to the same conclusion. Context has a huge part to play in how damaging poverty is, and affluent conditions are the most vulnerable to its academic effects.

Bayesian Posterior Estimates — Urbanicity Interaction Model

Parameter	Post. Mean	Post. SD	2.5%	97.5%
Family Income	0.313	0.014	0.285	0.340
Neighborhood Poverty (City)	-2.245	0.068	-2.378	-2.111
Urbanicity: Suburb	-0.015	0.005	-0.025	-0.005
Urbanicity: Town	0.025	0.005	0.015	0.035
Urbanicity: Rural	0.002	0.004	-0.006	0.010
Poverty $\times$ City	0.596	0.093	0.412	0.775
Poverty $\times$ Suburb	-1.154	0.067	-1.285	-1.021
Poverty $\times$ Town	0.499	0.069	0.363	0.634
Poverty $\times$ Rural	0.060	0.056	-0.051	0.171

Note. The Suburban interaction credible interval $[-1.285, -1.021]$ is entirely negative, confirming that suburban districts suffer a significantly steeper academic penalty from neighborhood poverty than city districts. The Rural interaction credible interval $[-0.051, 0.171]$ crosses zero, indicating the rural penalty, while present in the frequentist model, carries meaningful uncertainty in the Bayesian framework.

Dataset Completeness and Limitations

This dataset was a complete one as it covered nearly 450 million test scores across 11 years for most of the districts in the United States. However, there were several limitations. First, the dataset stops at 2019 and therefore was before the COVID-19 pandemic. The analysis is,

therefore, representative of pre-pandemic conditions, post pandemic analysis would be needed to assess whether neighborhood poverty effects increased after the pandemic.

Second, the analysis is conducted at district level instead of student level. District level rates, mask children from low-income families in wealthy areas and vice versa. Having information on an individual level would have created a more precise analysis.

Third, the dataset lacks other neighborhood characteristics such as crime rates, airquality, access to healthcare and food etc.. These would have added more insight into neighborhood factors that influenced test scores, and could lead to more targeted policy recommendations.

Recommendations to the State Education Commissioner

Priority 1: Adopt a Neighborhood-Level Resource Allocation Framework

Schools are currently funded with the focus mainly on poor individual families. However, this analysis shows that the neighborhood a child is in has an independent effect on academic outcomes. The commissioner should use a funding approach that accounts for both family poverty and neighborhood poverty allocation of resources. This would allow districts with deeply impoverished communities to receive resources that fit their demand. The interaction analysis makes this recommendation more crucial. Poverty does not do equal damage everywhere. High-income districts and suburban districts with unexpected poverty conditions show the steepest academic decline per unit poverty. This suggests that these conditions should be more scrutinized.

Priority 2: Focus Especially on Rural High-Poverty Districts

Rural districts make up over half of the districts in the dataset, and within this group there is a large variation. Some are doing well and others are deeply impoverished and invisible in national

policy conversations that are usually focused on urban school reform. The commissioner could create a formal Rural High Poverty designation for the most distressed rural and town districts and direct targeted funding, teacher incentives and infrastructure investments just for them.

Priority 3: Community Cultural Shift

The belief that academic outcomes are mostly based on individual effort and family responsibility has just been disproven. However, this consensus remains a widespread misconception. This analysis shows that the neighborhood has a significant part to play, as no individual effort can compensate for the compounding disadvantages of concentrated poverty. The commissioner should use these findings to explain in front of school boards, parent groups and local officials that neighborhood investment is a supplement to individual effort.

Conclusion

The evidence from this is clear. Neighborhood poverty is a strong and independent predator of academic outcomes in U.S. school districts. A 10 % point increase in neighborhood child poverty is associated with roughly a fifth of a standard deviation decrease in test scores after accounting for family income and district type. This was confirmed by our Bayesian analysis with 95% certainty that this effect is real. The model explains over 50% of variance in test scores, our proxy to academic performance, across districts, and neighborhood poverty carries more than five times the weight of family income. The conclusion is not that family income does not matter, but neighborhood conditions do as well.

Interaction analyses show a different dimension as it demonstrates the impact of neighborhood poverty is not constant in every condition. High-income districts and suburban districts suffer the most, which challenges the assumption that wealthy or suburban districts are not affected by poverty.

For a commissioner who is deciding how to allocate resources, neighborhoods are a prime avenue as well as individual families. Further research should use post-pandemic data, individual level records and more informative neighborhood characteristics that could provide a deeper picture.

Additional Resources

Data Source

Reardon, S. F., Ho, A. D., Shear, B. R., Fahle, E. M., Kalogrides, D., Jang, H., & Chavez, B. (2021). *Stanford Education Data Archive (Version 6.0)*. Retrieved from <https://edopportunity.org/opportunity/data/downloads/>

Literature

Reardon, S. F. (2011). The widening academic achievement gap between the rich and the poor: New evidence and possible explanations. In G. J. Duncan & R. J. Murnane (Eds.), *Whither opportunity? Rising inequality, schools, and children's life chances* (pp. 91–116). Russell Sage Foundation.

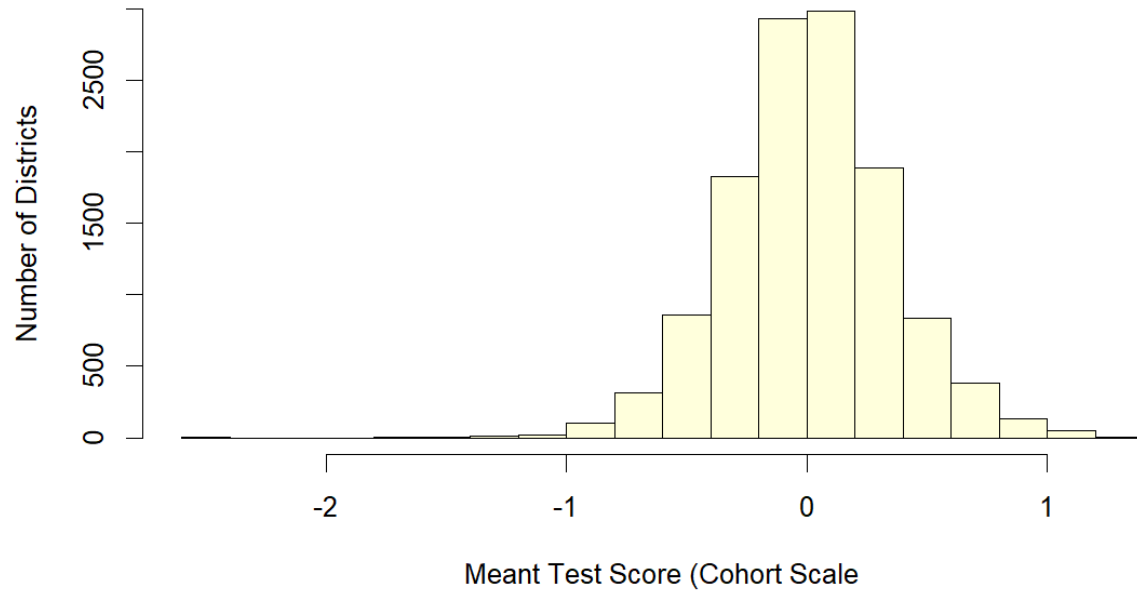
Chetty, R., Hendren, N., & Katz, L. F. (2016). The effects of exposure to better neighborhoods on children: New evidence from the Moving to Opportunity experiment. *American Economic Review*, 106(4), 855–902. <https://doi.org/10.1257/aer.20150572>

AI Assistance

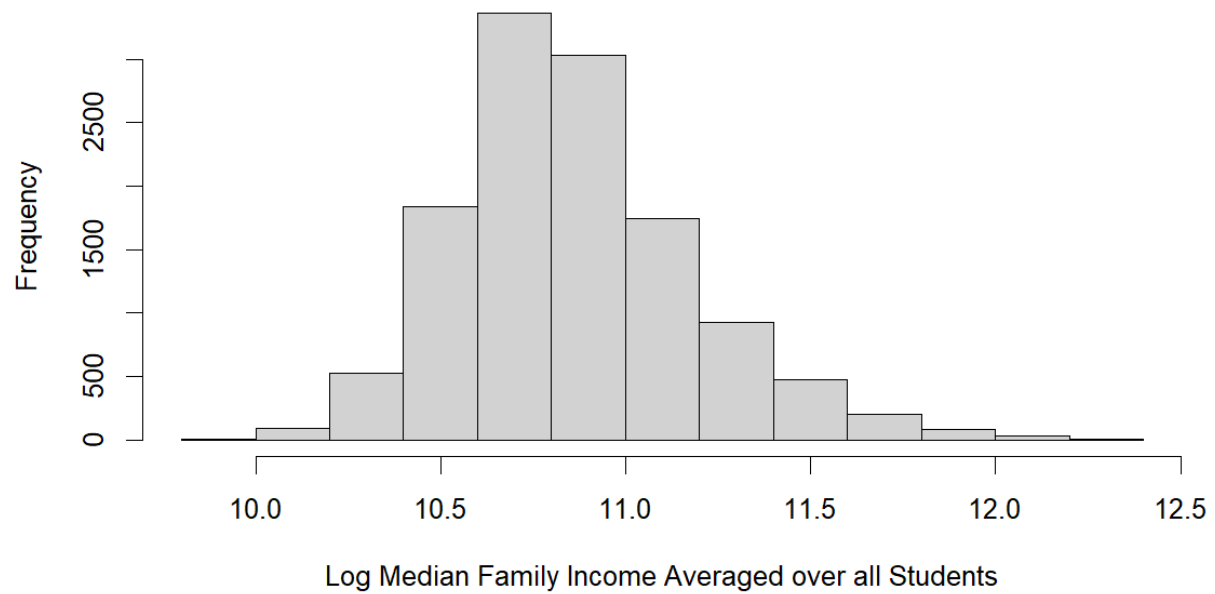
Anthropic. (2026). *Claude (claude-sonnet-4-6)* [Large language model]. Used for code development and writing support throughout this project. <https://www.anthropic.com>

Appendices

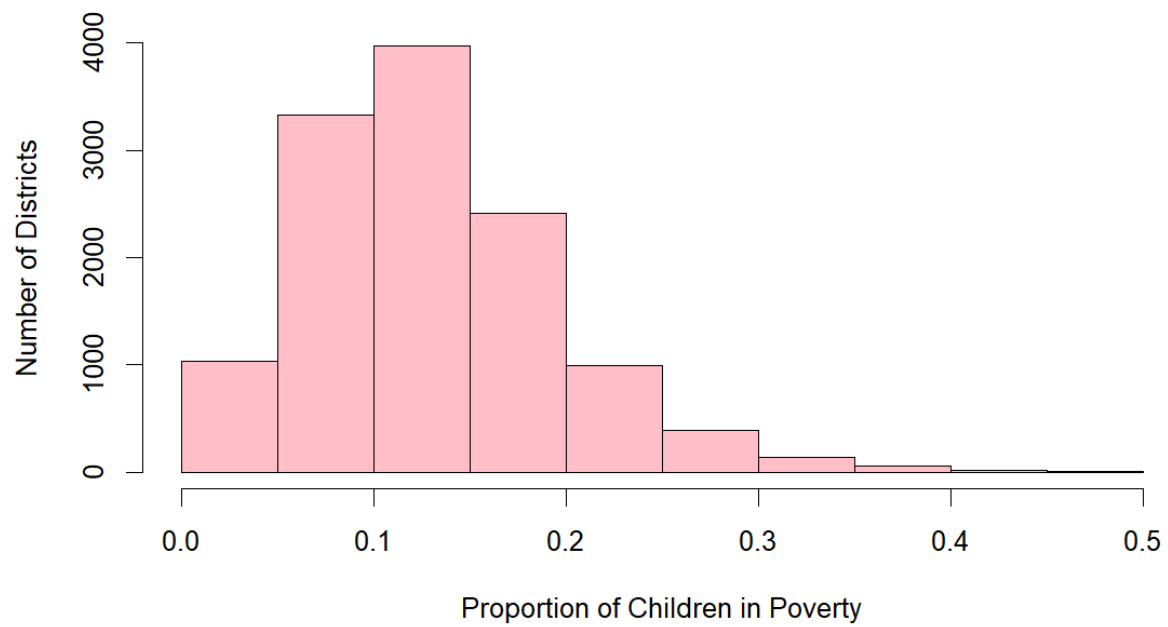
Distribution of Average Test Scores by District



Distirbution of Median Household Income by District



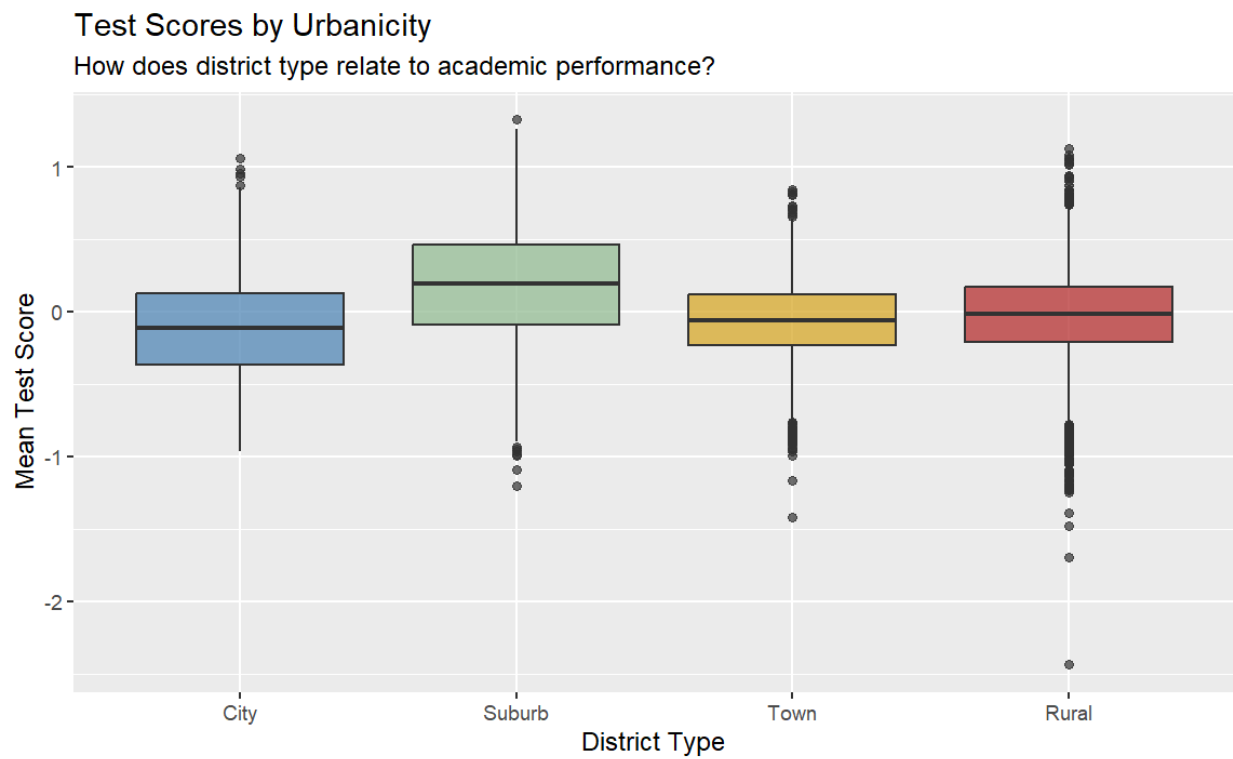
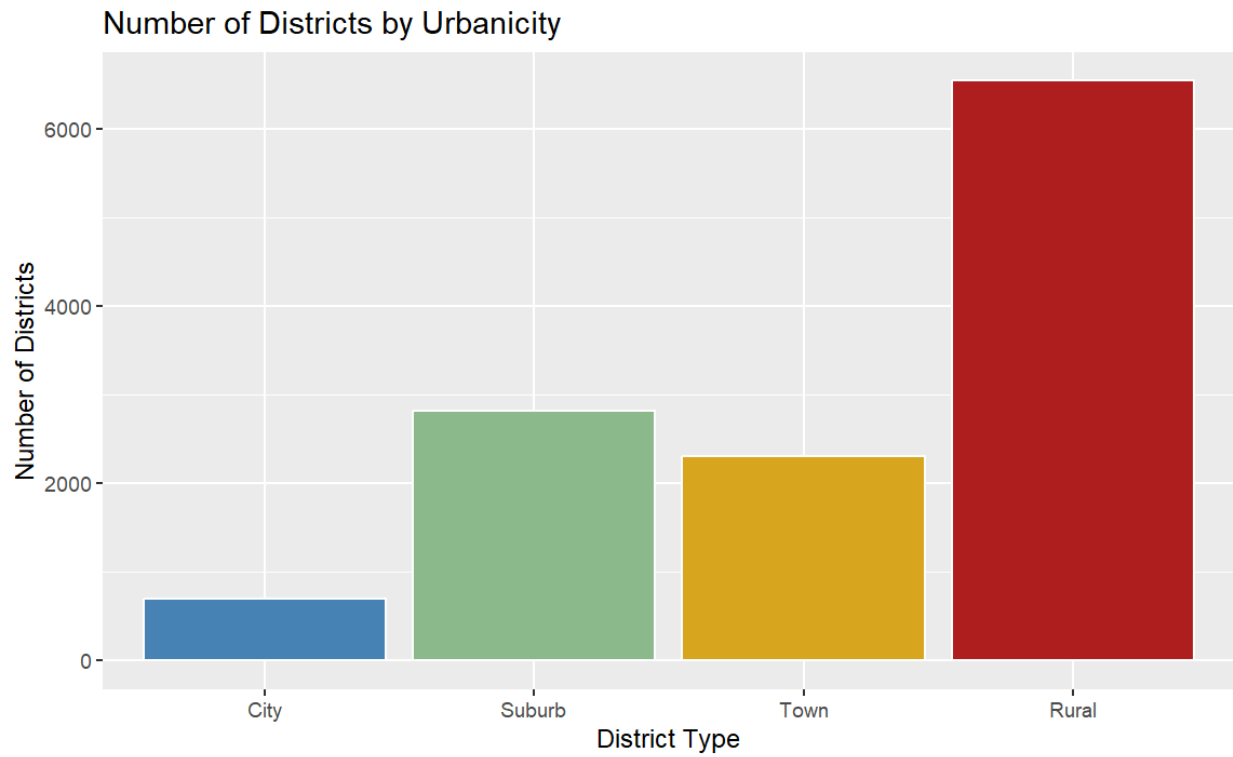
Distribution of Child Poverty over Districts



Family Income vs Neighborhood Poverty

How Correlated are the two predictors





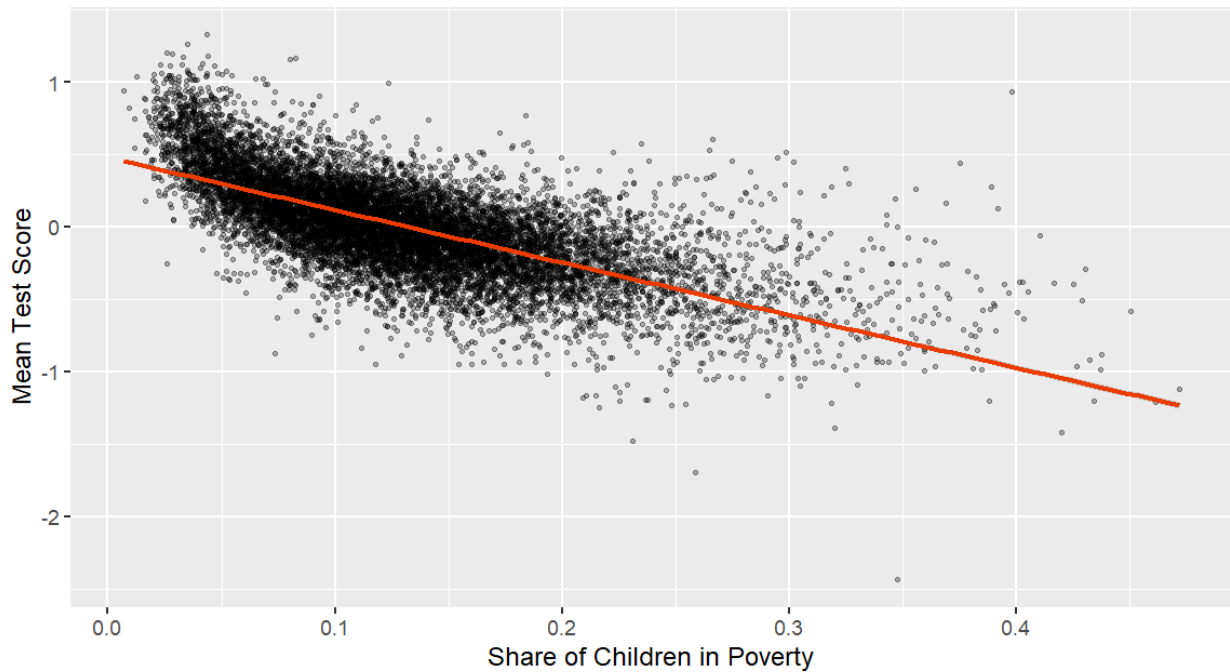
Family Income vs. Test Scores

The conventional story



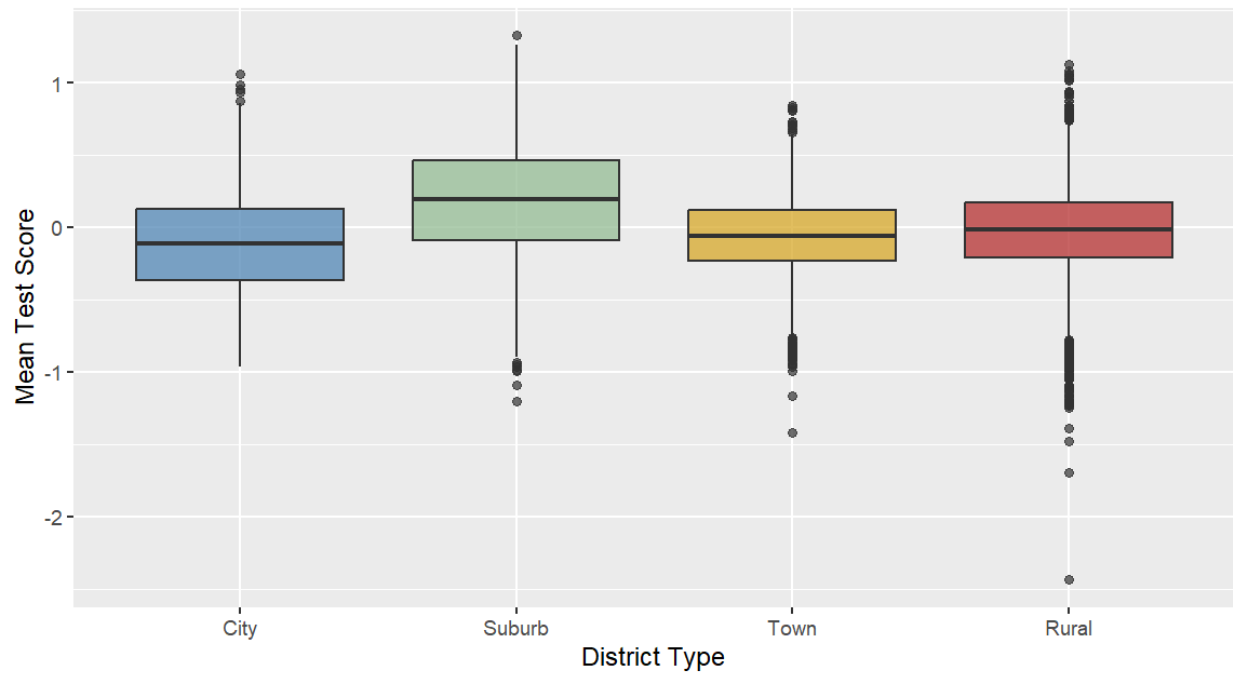
Neighborhood Poverty Rate vs. Test Scores

The neighborhood story



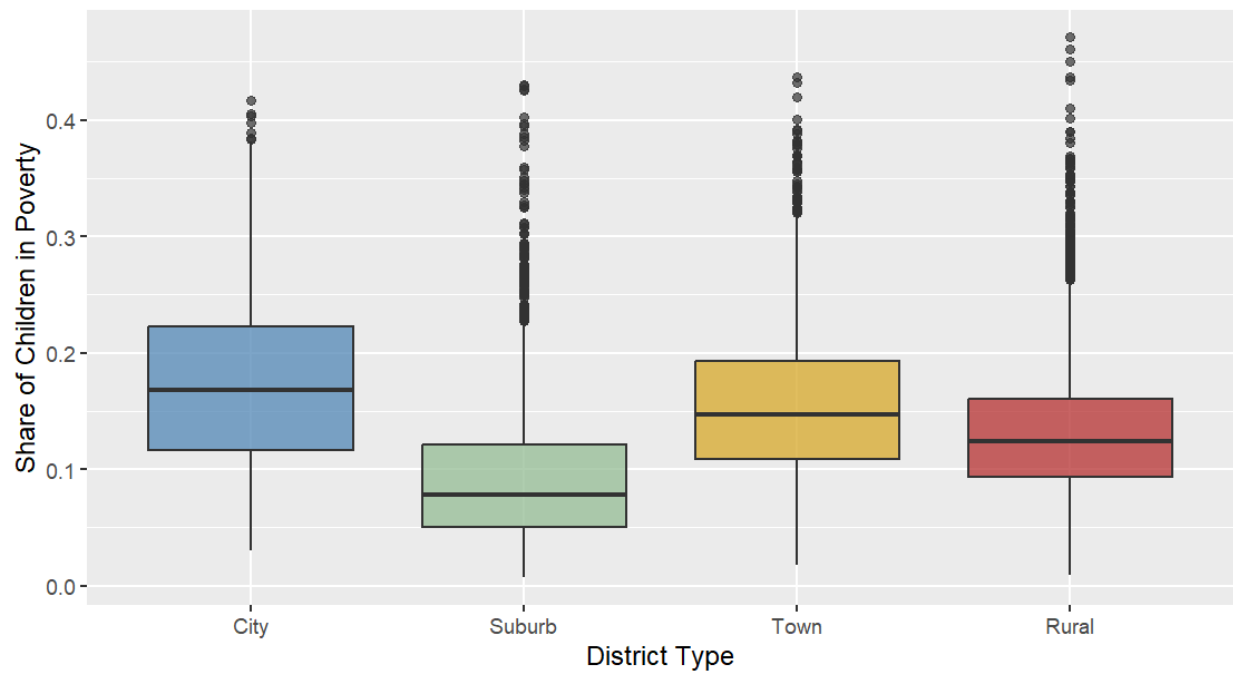
Test Scores by Urbanicity

How does district type relate to academic performance?



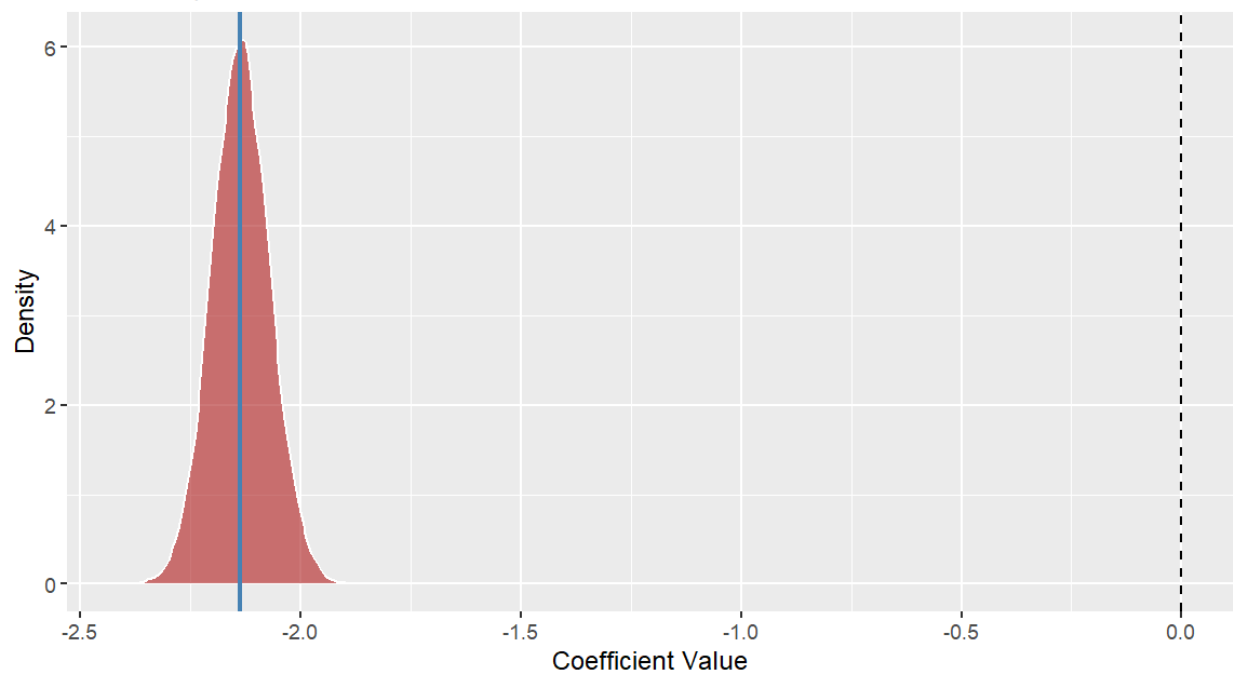
Neighborhood Poverty by Urbanicity

Do poverty rates differ across district types?



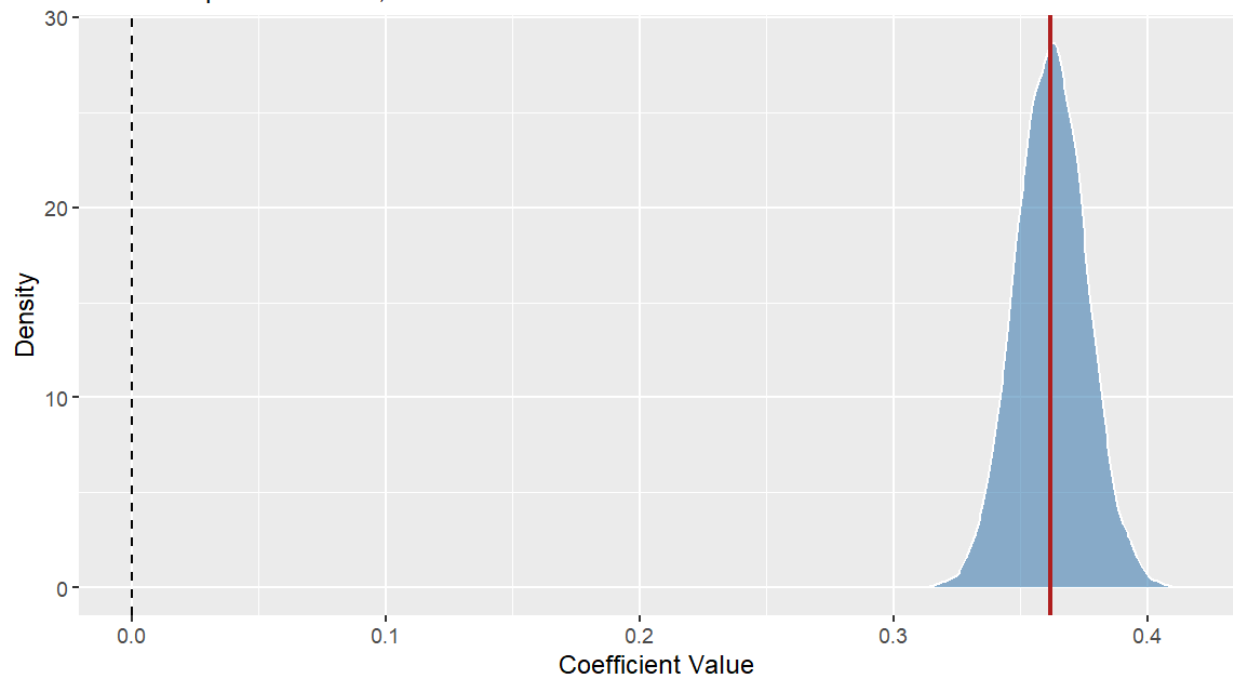
Posterior Distribution — Neighborhood Poverty Effect

Red line = posterior mean, dashed = zero



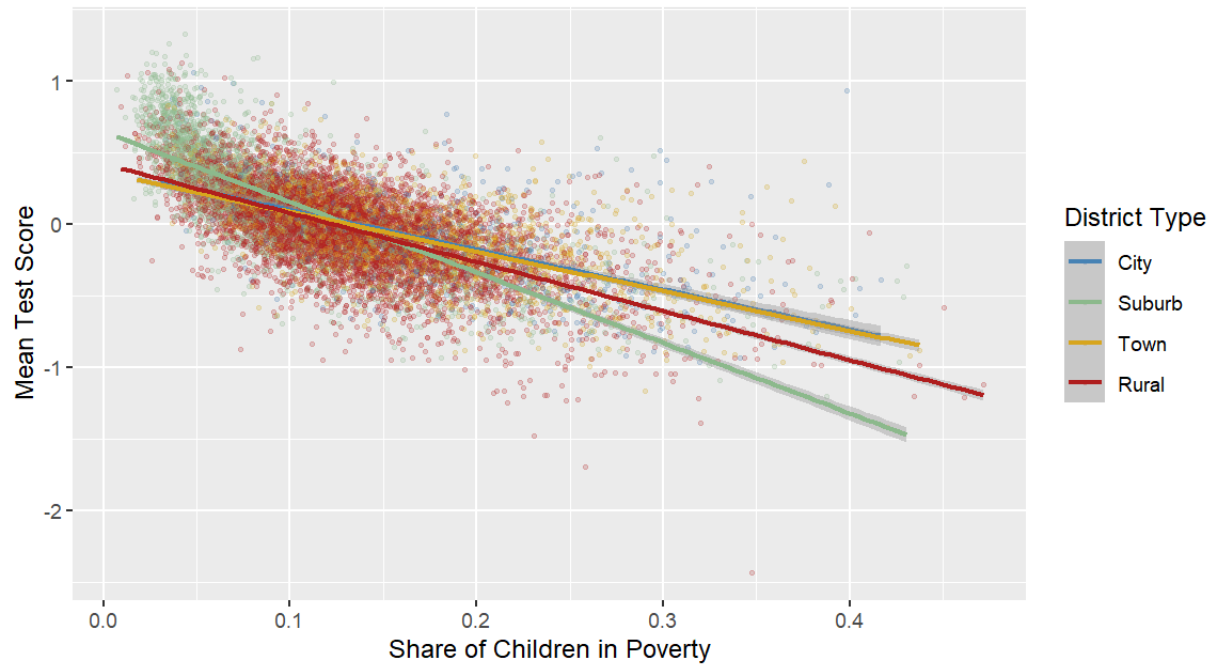
Posterior Distribution — Family Income Effect

Red line = posterior mean, dashed = zero



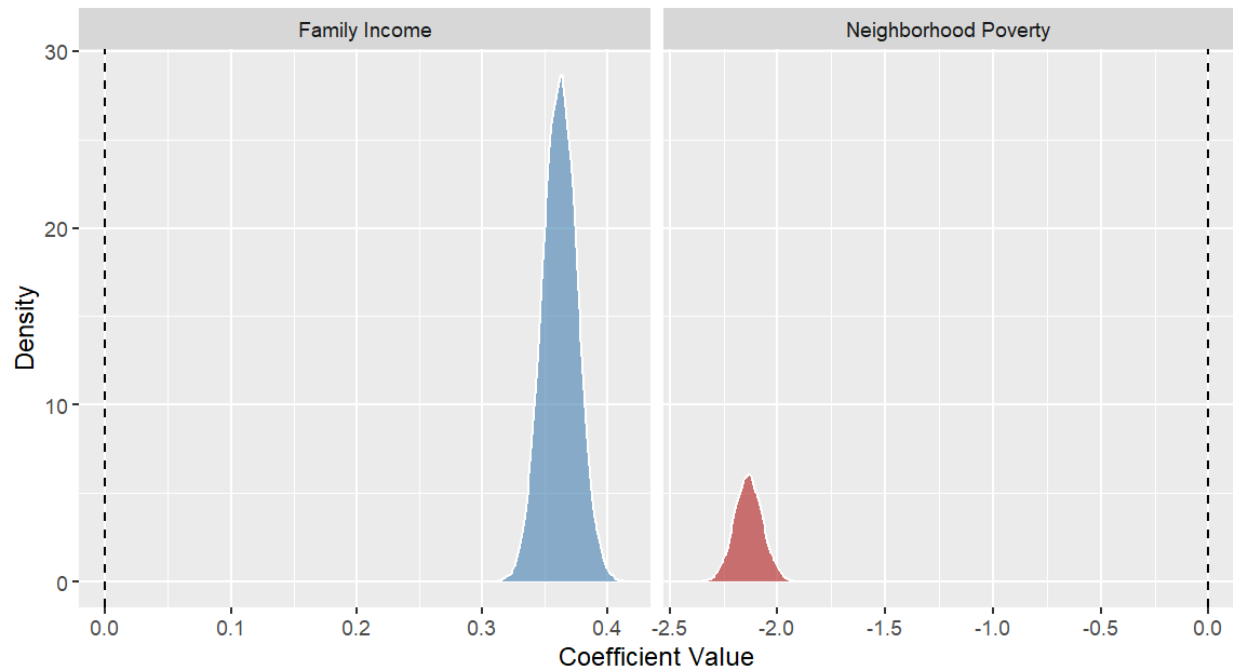
Neighborhood Poverty vs. Test Scores by District Type

Different slopes = interaction effect



Posterior Distributions of Predictor Effects

Dashed line = zero effect



Does Neighborhood Poverty Hurt Low-Income Districts More?

Steeper slope for low income = interaction effect

